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ROLL NO : 23I-0058

SECTION : BSE – 4A

OS ASSIGNMENT 01
QUESTION 02 REPORT

BSSE SPRING 2026

REPORT

1. SCRIPT OVERVIEW

File: server_monitor.sh

Purpose: Automated server resource monitoring with alerts

Features:

- ☒ Disk usage monitoring (alert >50%)
 - ☒ CPU usage monitoring (alert >50%)
 - ☒ Memory monitoring (alert if free <30%)
 - ☒ Log rotation (auto-rotate logs >25MB)
 - ☒ Comprehensive logging
 - ☒ Report generation
-

2. IMPLEMENTATION STEPS

Step 1: Script Setup

```
bash
#!/bin/bash
# Define thresholds and file paths
THRESHOLD_DISK=50
THRESHOLD_CPU=50
THRESHOLD_MEM=30
LOG_FILE="/var/log/server_monitor.log"
```

Step 2: Logging Function

```
bash
log_message() {
    echo "$(date) - $1" | sudo tee -a "$LOG_FILE"
}
```

Centralizes all logging with timestamps.

Step 3: Disk Monitoring

```
bash
check_disk_usage() {
    USAGE=$(df -h / | awk 'NR==2 {print $5}' | sed 's/%//')
    if [ $USAGE -gt $THRESHOLD_DISK ]; then
        echo "⚠ ALERT: Disk usage $USAGE%"
        log_message "ALERT: High disk usage"
    fi
}
```

Commands: df, awk, sed

Step 4: CPU Monitoring

```
bash
check_cpu_usage() {
    CPU=$(top -bn1 | grep "%Cpu" | awk '{print 100 - $8}')
    if (( $(echo "$CPU > $THRESHOLD_CPU" | bc -l) )); then
        echo "⚠ ALERT: CPU usage $CPU%"
        log_message "ALERT: High CPU usage"
        ps aux --sort=-%cpu | head -5
    fi
}
```

Commands: top, bc, ps

Step 5: Memory Monitoring

```
bash
check_memory_usage() {
    FREE=$(free -m | awk 'NR==2 {print ($4/$2)*100}')
    if (( $(echo "$FREE < $THRESHOLD_MEM" | bc -l) )); then
        echo "ALERT: Low memory ($FREE% free)"
        log_message "ALERT: Low memory"
        ps aux --sort=-%mem | head -5
    fi
}
```

Commands: free, bc, ps

Step 6: Log Rotation

bash

```
rotate_logs() {  
    if [ -f "$WATCH_LOG" ]; then  
        SIZE=$(sudo du -m "$WATCH_LOG" | cut -f1)  
        if [ $SIZE -gt $MAX_LOG_SIZE ]; then  
            BACKUP="/var/log/backups/app_$(date +%Y%m%d_%H%M%S).log"  
            sudo mkdir -p /var/log/backups  
            sudo cp "$WATCH_LOG" "$BACKUP"  
            sudo gzip "$BACKUP"  
            sudo truncate -s 0 "$WATCH_LOG"  
            log_message "Log rotated: $BACKUP.gz"  
        fi  
    fi  
}
```

Commands: du, cp, gzip, truncate

Step 7: Report Generation

bash

```
generate_report() {  
    REPORT="/tmp/report_$(date +%Y%m%d_%H%M%S).txt"  
    {  
        echo "=== SERVER REPORT ==="  
        echo "Host: $(hostname)"  
        echo "Kernel: $(uname -r)"  
        echo "Uptime: $(uptime -p)"  
        df -h /  
        free -h  
        ps aux --sort=-%cpu | head -5  
    } > "$REPORT"  
    echo "Report: $REPORT"  
}
```

Step 8: Error Handling

```
bash
handle_error() {
    echo "ERROR: $1"
    log_message "ERROR: $1"
}
```

Step 9: Main Function

```
bash
main() {
    echo "SERVER MONITOR - Shah Faisal (23I-0058)"
    echo "Start: $(date)"

    check_disk_usage || handle_error "Disk check failed"
    check_cpu_usage || handle_error "CPU check failed"
    check_memory_usage || handle_error "Memory check failed"
    rotate_logs || handle_error "Log rotation failed"
    generate_report

    echo "Completed: $(date)"
    log_message "Monitoring completed"
}

main
```

```
shah@DESKTOP-02Q2MPC:~$ cat /var/log/server_monitor.log
2026-02-22 12:09:25 - Server Monitor Started
2026-02-22 12:09:25 - Checking Disk Usage...
2026-02-22 12:09:25 - Disk Usage: 1%
2026-02-22 12:09:25 - Disk usage normal
2026-02-22 12:09:25 - Checking CPU Usage...
2026-02-22 12:09:25 - CPU Usage: 100%
2026-02-22 12:09:25 - ALERT: High CPU usage - 100%
2026-02-22 12:09:25 - Checking Memory Usage...
2026-02-22 12:09:25 - Memory - Total: 7875MB, Free: 86%
2026-02-22 12:09:25 - Memory usage normal
2026-02-22 12:09:25 - Checking Log Rotation...
2026-02-22 12:09:25 - Log size: 1MB
2026-02-22 12:09:25 - Report generated: /tmp/server_report_20260222_120926.txt
2026-02-22 12:09:25 - Server monitoring completed
2026-02-22 12:12:07 - Server Monitor Started
2026-02-22 12:12:07 - Checking Disk Usage...
2026-02-22 12:12:07 - Disk Usage: 1%
2026-02-22 12:12:07 - Disk usage normal
2026-02-22 12:12:07 - Checking CPU Usage...
2026-02-22 12:12:07 - CPU Usage: 100%
2026-02-22 12:12:07 - ALERT: High CPU usage - 100%
2026-02-22 12:12:07 - Checking Memory Usage...
2026-02-22 12:12:07 - Memory - Total: 7875MB, Free: 85%
2026-02-22 12:12:07 - Memory usage normal
2026-02-22 12:12:07 - Checking Log Rotation...
2026-02-22 12:12:07 - Log size: 1MB
2026-02-22 12:12:07 - Report generated: /tmp/server_report_20260222_121208.txt
2026-02-22 12:12:07 - Server monitoring completed
shah@DESKTOP-02Q2MPC:~$
```

```
shah@DESKTOP-02Q2MPC: ~
shah@DESKTOP-02Q2MPC:~$ cat /tmp/server_report_*.txt
=====
SERVER MONITORING REPORT
Generated: Sun Feb 22 12:09:26 PKT 2026
=====

SYSTEM INFORMATION:
-----
Hostname: DESKTOP-02Q2MPC
Kernel: 6.6.87.2-microsoft-standard-WSL2
Uptime: up 4 minutes

DISK USAGE:
-----
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdd        1007G  1.5G  955G   1% /

CPU INFORMATION:
-----
CPU Usage: %Cpu(s):  0.0 us,  1.6 sy,  0.0 ni, 98.4 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
Load Avg:  0.20, 0.06, 0.02

MEMORY INFORMATION:
-----
              total        used        free      shared  buff/cache   available
Mem:          7.7Gi        304Mi        6.6Gi         3.0Mi         767Mi         7.2Gi
Swap:         2.0Gi           0B         2.0Gi

TOP PROCESSES:
-----
By CPU:
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
shah      1904  4.0  0.0   5048   3712 pts/0    S+   12:09   0:00 /bin/bash ./server_monitor.sh
root         1  0.3  0.1 166068 10856 ?        Ss   12:05   0:00 /sbin/init
shah      280  0.3  0.0   6208   4992 pts/0    Ss   12:05   0:00 -bash
root     1807  0.3  0.2 293092 20096 ?        Ssl  12:09   0:00 /usr/libexec/packagekitd

By Memory:
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root      225  0.0  0.2 107164 20992 ?        Ssl  12:05   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutd
own --wait-for-signal
root     1807  0.3  0.2 293092 20096 ?        Ssl  12:09   0:00 /usr/libexec/packagekitd
root      187  0.0  0.2 29112 18676 ?        Ss   12:05   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
root        62  0.2  0.1 39672 15232 ?        S<s  12:05   0:00 /lib/systemd/systemd-journald
```

```
shah@DESKTOP-02Q2MPC: ~
LOG STATUS:
-----
Monitor Log: /var/log/server_monitor.log
Watch Log: /var/log/application.log (12K )
=====
SERVER MONITORING REPORT
Generated: Sun Feb 22 12:12:08 PKT 2026
=====

SYSTEM INFORMATION:
-----
Hostname: DESKTOP-02Q2MPC
Kernel: 6.6.87.2-microsoft-standard-WSL2
Uptime: up 7 minutes

DISK USAGE:
-----
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdd         1007G  1.5G  955G   1% /

CPU INFORMATION:
-----
CPU Usage: %Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
Load Avg:  0.01, 0.03, 0.00

MEMORY INFORMATION:
-----
Mem:            total      used      free      shared  buff/cache  available
Swap:           2.0Gi       0B       2.0Gi       3.0Mi       768Mi       7.2Gi

TOP PROCESSES:
-----
By CPU:
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.2  0.1 166068 10856 ?        Ss   12:05   0:00 /sbin/init
root        62  0.1  0.1  47868 15360 ?        S<s  12:05   0:00 /lib/systemd/systemd-journald
shah       280  0.1  0.0   6208  4992 pts/0    Ss   12:05   0:00 -bash
root         2  0.0  0.0   3120  2048 ?        Sl   12:05   0:00 /init

By Memory:
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root       225  0.0  0.2 107164 20992 ?        Ssl  12:05   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutd
own --wait-for-signal

shah@DESKTOP-02Q2MPC:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G   0    3.9G   0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none            3.9G  4.0K   3.9G   1% /mnt/wsl
drivers         113G  107G   6.1G  95% /usr/lib/wsl/drivers
/dev/sdd        1007G  1.5G  955G   1% /
none            3.9G   80K   3.9G   1% /mnt/wslg
none            3.9G   0    3.9G   0% /usr/lib/wsl/lib
rootfs          3.9G  2.7M   3.9G   1% /init
none            3.9G  516K   3.9G   1% /run
none            3.9G   0    3.9G   0% /run/lock
none            3.9G   0    3.9G   0% /run/shm
none            3.9G   76K   3.9G   1% /mnt/wslg/versions.txt
none            3.9G   76K   3.9G   1% /mnt/wslg/doc
C:\             113G  107G   6.1G  95% /mnt/c
E:\             113G   29G   84G  26% /mnt/e
tmpfs           788M   8.0K  788M   1% /run/user/1000
shah@DESKTOP-02Q2MPC:~$
```

```
shah@DESKTOP-02Q2MPC:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:          7.7Gi        341Mi        7.3Gi         3.0Mi         99Mi        7.2Gi
Swap:         2.0Gi           0B         2.0Gi
shah@DESKTOP-02Q2MPC:~$
```

```
shah@DESKTOP-02Q2MPC:~$ top -bn1 | head -15
top - 12:16:17 up 10 min, 1 user, load average: 0.03, 0.02, 0.00
Tasks: 27 total, 1 running, 26 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7875.2 total, 7428.7 free, 346.1 used, 100.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7395.5 avail Mem

  PID USER      PR  NI   VIRT   RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0 166068  10856  8168 S   0.0   0.1   0:00.87 systemd
    2 root        20   0   3120   2048  1920 S   0.0   0.0   0:00.01 init-systemd(Ub
    7 root        20   0   3576   2296  1792 S   0.0   0.0   0:00.34 init
   62 root        19  -1  47868 15360 14592 S   0.0   0.2   0:00.75 systemd-journal
   86 root        20   0  22972   5760  4608 S   0.0   0.1   0:00.41 systemd-udev
   94 systemd+    20   0  89364   7168  6400 S   0.0   0.1   0:00.31 systemd-timesyn
  102 systemd+    20   0  16128   8064  7168 S   0.0   0.1   0:00.12 systemd-network
  180 root        20   0   4308   2688  2432 S   0.0   0.0   0:00.00 cron
shah@DESKTOP-02Q2MPC:~$
```